

Geometry & Measurement

ANSWER KEY



Angle facts

- 1 Two angles on a straight line measure $3x$ and $2x + 30$. Find x .

Angles on a straight line sum to 180° : $3x + 2x + 30 = 180 \Rightarrow 5x = 150 \Rightarrow x = 30$.

- 2 Find the angle vertically opposite a 65° angle.

Vertically opposite angles are equal, so the angle is 65° .

- 3 Three angles around a point measure 110° , 95° , and x . Find x .

Angles around a point sum to 360° : $x = 360 - 110 - 95 = 155^\circ$.

- 4 Two angles are complementary (sum to 90°). One measures 37° . Find the other.

$90 - 37 = 53^\circ$.

Parallel lines and transversals

- 5 Two parallel lines are cut by a transversal, creating an angle of 70° . Find its corresponding angle.

Corresponding angles are equal, so the angle is 70° .

- 6 In the same setup, find the co-interior (allied) angle to a 70° angle.

Co-interior angles sum to 180° : $180 - 70 = 110^\circ$.

- 7 Find the alternate interior angle to a 55° angle.

Alternate interior angles are equal, so the angle is 55° .

- 8 Two parallel lines cut by a transversal create an angle of $4x$ and its co-interior angle of $2x + 30$. Find x .

Co-interior angles sum to 180° : $4x + 2x + 30 = 180 \Rightarrow 6x = 150 \Rightarrow x = 25$.

Triangle angle sum

- 9 Find the third angle of the triangle shown, given the two marked angles.

$180 - 40 - 75 = 65^\circ$.

- 10** An isosceles triangle has two equal angles of 72° each. Find the third angle.
- $$180 - 72 - 72 = 36^\circ.$$
- 11** In a right triangle, one of the non-right angles is 35° . Find the other non-right angle.
- $$180 - 90 - 35 = 55^\circ.$$
- 12** A triangle has three equal angles. Find each angle, and name this type of triangle.
- $$180 \div 3 = 60^\circ \text{ each — this is an equilateral (equiangular) triangle.}$$
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Angle sums of polygons

- 13** Find the sum of the interior angles of a hexagon (6 sides).
- $$\text{Sum} = (n - 2) \times 180^\circ = (6 - 2) \times 180^\circ = 720^\circ.$$
- 14** Find the sum of the interior angles of a pentagon (5 sides).
- $$\text{Sum} = (5 - 2) \times 180^\circ = 540^\circ.$$
- 15** A regular polygon has an interior angle sum of $1,080^\circ$. How many sides does it have?
- $$(n - 2) \times 180 = 1,080 \Rightarrow n - 2 = 6 \Rightarrow n = 8 \text{ sides (a regular octagon).}$$
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The Pythagorean theorem

- 16** Find the hypotenuse of the right triangle shown.
- $$c = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10 \text{ cm.}$$
- 17** A right triangle has hypotenuse 13 and one leg 5. Find the other leg.
- $$\sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12.$$
- 18** A 10 m ladder leans against a wall with its foot 6 m from the wall's base. How high up the wall does it reach?
- $$\sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = 8 \text{ m.}$$
- 19** A right triangle has legs 9 and 12. Find the hypotenuse.
- $$\sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15.$$
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Perimeter and area of rectangles and squares

- 20** A rectangle has length 12 cm and width 7 cm. Find its perimeter and area.

$$\text{Perimeter} = 2(12 + 7) = 38 \text{ cm. Area} = 12 \times 7 = 84 \text{ cm}^2.$$

- 21** A square has side length 9 cm. Find its perimeter and area.

$$\text{Perimeter} = 4 \times 9 = 36 \text{ cm. Area} = 9^2 = 81 \text{ cm}^2.$$

- 22** A rectangle has perimeter 50 cm and length 16 cm. Find its width and area.

$$\text{Width} = \frac{50}{2} - 16 = 25 - 16 = 9 \text{ cm. Area} = 16 \times 9 = 144 \text{ cm}^2.$$

- 23** A square has area 121 cm^2 . Find its side length and perimeter.

$$\text{Side} = \sqrt{121} = 11 \text{ cm. Perimeter} = 4 \times 11 = 44 \text{ cm.}$$

Area of triangles and parallelograms

- 24** A triangle has base 14 cm and height 9 cm. Find its area.

$$\text{Area} = \frac{1}{2} \times 14 \times 9 = 63 \text{ cm}^2.$$

- 25** A parallelogram has base 12 cm and height 8 cm. Find its area.

$$\text{Area} = 12 \times 8 = 96 \text{ cm}^2.$$

- 26** A triangle has area 60 cm^2 and base 15 cm. Find its height.

$$\text{Height} = \frac{2 \times 60}{15} = 8 \text{ cm.}$$

- 27** A parallelogram has area 126 cm^2 and height 9 cm. Find its base.

$$\text{Base} = \frac{126}{9} = 14 \text{ cm.}$$

Circles: circumference and area

- 28** A circle has radius 7 cm. Find its circumference, using $\pi = \frac{22}{7}$.

$$C = 2\pi r = 2 \times \frac{22}{7} \times 7 = 44 \text{ cm.}$$

- 29** A circle has radius 7 cm. Find its area, using $\pi = \frac{22}{7}$.

$$A = \pi r^2 = \frac{22}{7} \times 49 = 154 \text{ cm}^2.$$

- 30** A circle has diameter 20 cm. Find its circumference in terms of π .

$$C = \pi d = 20\pi \text{ cm.}$$

- 31 Find the area of the shaded 90° sector shown, in terms of π .

$$\text{Sector area} = \frac{90}{360} \times \pi \times 10^2 = \frac{1}{4} \times 100\pi = 25\pi \text{ cm}^2.$$

Composite figures

- 32 Find the area of the L-shaped room shown.

Treat it as a $10 \text{ m} \times 8 \text{ m}$ rectangle with a $4 \text{ m} \times 3 \text{ m}$ notch removed: $10 \times 8 - 4 \times 3 = 80 - 12 = 68 \text{ m}^2$.

- 33 A garden is L-shaped: a 15 m by 6 m rectangle with a 5 m by 2 m rectangle cut from one corner. Find its area.

$$15 \times 6 - 5 \times 2 = 90 - 10 = 80 \text{ m}^2.$$

- 34 A shape consists of a 12 cm by 5 cm rectangle with a right triangle of base 6 cm and height 5 cm attached to one of the short sides. Find the total area.

$$\text{Rectangle: } 12 \times 5 = 60 \text{ cm}^2. \text{ Triangle: } \frac{1}{2} \times 6 \times 5 = 15 \text{ cm}^2. \text{ Total} = 60 + 15 = 75 \text{ cm}^2.$$

Volume and surface area

- 35 A rectangular box has dimensions $5 \text{ cm} \times 4 \text{ cm} \times 3 \text{ cm}$. Find its volume.

$$V = 5 \times 4 \times 3 = 60 \text{ cm}^3.$$

- 36 A cube has edge length 6 cm . Find its volume and surface area.

$$V = 6^3 = 216 \text{ cm}^3. \text{ Surface area} = 6 \times 6^2 = 6 \times 36 = 216 \text{ cm}^2.$$

- 37 A rectangular prism has volume 240 cm^3 , length 10 cm , and width 6 cm . Find its height.

$$\text{Height} = \frac{240}{10 \times 6} = \frac{240}{60} = 4 \text{ cm}.$$

- 38 Find the surface area of a rectangular box with dimensions $8 \text{ cm} \times 5 \text{ cm} \times 4 \text{ cm}$.

$$SA = 2(lw + lh + wh) = 2(40 + 32 + 20) = 2 \times 92 = 184 \text{ cm}^2.$$

Word problems

- 39 A rectangular garden measuring 20 m by 15 m is surrounded by a path 1 m wide. Find the area of the path alone.

Outer dimensions: $22 \text{ m} \times 17 \text{ m} = 374 \text{ m}^2$. Garden area: $20 \times 15 = 300 \text{ m}^2$. Path area
 $= 374 - 300 = 74 \text{ m}^2$.

- 40 New flooring costs 45 riyals per square meter. Find the total cost to floor a rectangular room measuring 6 m by 4 m.

Area $= 6 \times 4 = 24 \text{ m}^2$. Cost $= 24 \times 45 = 1,080$ riyals.

- 41 A 17 m ladder leans against a vertical wall with its base 8 m from the wall. How high up the wall does it reach?

$$\sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = 15 \text{ m}.$$

- 42 A jogger runs 4 laps of a circular track of radius 35 m, using $\pi = \frac{22}{7}$. Find the total distance covered.

One lap $= 2\pi r = 2 \times \frac{22}{7} \times 35 = 220 \text{ m}$. Four laps $= 4 \times 220 = 880 \text{ m}$.

MCQ — Angle facts

- 43 Two angles are supplementary. One is three times the other. Find the smaller angle.

- a. 60°
- b. 30°
- c. 135°
- d. 45° ✓

$$x + 3x = 180 \Rightarrow x = 45^\circ.$$

- 44 Three angles around a point measure 118° , 142° , and x . Find x .

- a. 80°
- b. 140°
- c. 120°
- d. 100° ✓

$$x = 360 - 118 - 142 = 100^\circ.$$

45 An angle is four times its complement. Find the angle.

- a. 54°
- b. 18°
- c. 72° ✓
- d. 90°

$$x = 4(90 - x) \Rightarrow 5x = 360 \Rightarrow x = 72^\circ.$$

46 Two angles on a straight line measure $(4x + 10)^\circ$ and $(2x + 20)^\circ$. Find x .

- a. 20
- b. 35
- c. 15
- d. 25 ✓

$$4x + 10 + 2x + 20 = 180 \Rightarrow 6x + 30 = 180 \Rightarrow x = 25.$$

MCQ — Parallel lines and transversals

47 Two parallel lines are cut by a transversal. One angle is $(3x + 15)^\circ$ and its corresponding angle is $(5x - 25)^\circ$. Find x .

- a. 10
- b. 8
- c. 30
- d. 20 ✓

$$\text{Corresponding angles are equal: } 3x + 15 = 5x - 25 \Rightarrow x = 20.$$

- 48 Two parallel lines cut by a transversal create co-interior angles of $(2x + 10)^\circ$ and $(3x - 30)^\circ$. Find x .
- a. 30
 - b. 40 ✓
 - c. 50
 - d. 34

Co-interior angles sum to 180° : $2x + 10 + 3x - 30 = 180 \Rightarrow 5x = 200 \Rightarrow x = 40$.

- 49 Two parallel lines are cut by a transversal. One angle is 63° . Find its alternate exterior angle.
- a. 153°
 - b. 27°
 - c. 63° ✓
 - d. 117°

Alternate exterior angles are equal, so the answer is 63° .

- 50 Two parallel lines are cut by a transversal. One angle and its co-interior angle differ by 40° and sum to 180° . Find the larger angle.
- a. 110° ✓
 - b. 70°
 - c. 140°
 - d. 90°

$$L + S = 180, L - S = 40 \Rightarrow L = 110^\circ.$$

MCQ — Triangle angle sum

51 Find the third angle of the triangle shown.

- a. 116°
- b. 64°
- c. 54°
- d. 44° ✓

$$180 - 64 - 72 = 44^\circ.$$

52 The isosceles triangle shown has two equal base angles of 70° each. Find the apex angle.

- a. 55°
- b. 70°
- c. 40° ✓
- d. 110°

$$180 - 70 - 70 = 40^\circ.$$

53 A triangle has three angles in the ratio 2 : 3 : 4. Find the largest angle.

- a. 90°
- b. 100°
- c. 80° ✓
- d. 60°

$$9 \text{ parts} = 180^\circ, \text{ so } 1 \text{ part} = 20^\circ; \text{ largest} = 4 \times 20 = 80^\circ.$$

54 In a right triangle, one non-right angle exceeds the other by 26° . Find the larger non-right angle.

- a. 32°
- b. 58° ✓
- c. 64°
- d. 48°

$$x + (x + 26) = 90 \Rightarrow x = 32; \text{ larger} = 58^\circ.$$

MCQ — The Pythagorean theorem

55 Find the hypotenuse of the right triangle shown.

- a. 23
- b. 19
- c. 17 ✓
- d. 13

$$\sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17.$$

56 Find the missing leg of the right triangle shown.

- a. 26
- b. 18
- c. 24 ✓
- d. 20

$$\sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24.$$

57 A rectangular field is 24 m long and 18 m wide. Find the length of its diagonal.

- a. 30 ✓
- b. 36
- c. 42
- d. 26

$$\sqrt{24^2 + 18^2} = \sqrt{576 + 324} = \sqrt{900} = 30 \text{ m.}$$

58 A triangle has sides 9 cm, 12 cm, and 15 cm, as shown. Is it a right triangle?

- a. Only if it is also isosceles
- b. Yes — a right triangle ✓
- c. Cannot be determined
- d. No — not a right triangle

$9^2 + 12^2 = 81 + 144 = 225 = 15^2$, so yes, it is a right triangle.

59 A 15 m ladder leans against a wall with its foot 9 m from the wall's base. How high up the wall does the ladder reach?

- a. 13
- b. 10
- c. 12 ✓
- d. 18

$\sqrt{15^2 - 9^2} = \sqrt{225 - 81} = \sqrt{144} = 12$ m.

MCQ — Perimeter and area of rectangles and squares

60 Find the perimeter and area of the rectangle shown.

- a. $P = 48$ cm, $A = 135$ cm² ✓
- b. $P = 48$ cm, $A = 24$ cm²
- c. $P = 135$ cm, $A = 48$ cm²
- d. $P = 24$ cm, $A = 135$ cm²

$P = 2(15 + 9) = 48$ cm; $A = 15 \times 9 = 135$ cm².

61 Find the perimeter and area of the square shown.

a. $P = 44 \text{ cm}, A = 110 \text{ cm}^2$

b. $P = 44 \text{ cm}, A = 121 \text{ cm}^2$ ✓

c. $P = 22 \text{ cm}, A = 121 \text{ cm}^2$

d. $P = 121 \text{ cm}, A = 44 \text{ cm}^2$

$$P = 4 \times 11 = 44 \text{ cm}; A = 11^2 = 121 \text{ cm}^2.$$

62 A rectangle has perimeter 64 cm and length 20 cm. Find its width and area.

a. $w = 12 \text{ cm}, A = 32 \text{ cm}^2$

b. $w = 12 \text{ cm}, A = 240 \text{ cm}^2$ ✓

c. $w = 44 \text{ cm}, A = 240 \text{ cm}^2$

d. $w = 22 \text{ cm}, A = 440 \text{ cm}^2$

$$2(l + w) = 64 \Rightarrow l + w = 32 \Rightarrow w = 12 \text{ cm}; A = 20 \times 12 = 240 \text{ cm}^2.$$

63 A square has area 225 cm^2 . Find its side length and perimeter.

a. $s = 15 \text{ cm}, P = 30 \text{ cm}$

b. $s = 225 \text{ cm}, P = 900 \text{ cm}$

c. $s = 15 \text{ cm}, P = 60 \text{ cm}$ ✓

d. $s = 45 \text{ cm}, P = 60 \text{ cm}$

$$s = \sqrt{225} = 15 \text{ cm}; P = 4 \times 15 = 60 \text{ cm}.$$

MCQ — Area of triangles and parallelograms

64 The triangle shown has base 16 cm and height 10 cm. Find its area.

a. 160

b. 26

c. 80 ✓

d. 40

$$A = \frac{1}{2}(16)(10) = 80 \text{ cm}^2.$$

65 The parallelogram shown has base 18 cm and height 7 cm. Find its area.

a. 252

b. 126 ✓

c. 63

d. 25

$$A = 18 \times 7 = 126 \text{ cm}^2.$$

66 A triangle has area 84 cm^2 and height 12 cm. Find its base.

a. 14 ✓

b. 1,008

c. 28

d. 7

$$b = \frac{2(84)}{12} = 14 \text{ cm}.$$

67 A parallelogram has area 150 cm^2 and base 15 cm. Find its height.

a. 20

b. 10 ✓

c. 2,250

d. 5

$$h = \frac{150}{15} = 10 \text{ cm}.$$

MCQ — Circles: circumference and area

68 Find the circumference of the circle shown, using $\pi = \frac{22}{7}$.

- a. 44
- b. 616
- c. 88 ✓
- d. 176

$$C = 2 \times \frac{22}{7} \times 14 = 88 \text{ cm.}$$

69 Find the area of the circle shown, using $\pi = \frac{22}{7}$.

- a. 132
- b. 2,772
- c. 693
- d. 1,386 ✓

$$A = \frac{22}{7} \times 21^2 = 1,386 \text{ cm}^2.$$

70 Find the circumference of the circle shown, in terms of π .

- a. 48π cm
- b. 24π cm ✓
- c. 12π cm
- d. 144π cm

$$C = 2\pi r = 2\pi(12) = 24\pi \text{ cm.}$$

71 Find the area of the shaded 60° sector shown, in terms of π .

- a. $81\pi \text{ cm}^2$
- b. $27\pi \text{ cm}^2$
- c. $13.5\pi \text{ cm}^2$ ✓
- d. $4.5\pi \text{ cm}^2$

$$\text{Sector area} = \frac{60}{360} \times \pi(9)^2 = 13.5\pi \text{ cm}^2.$$

72 A circle has area $144\pi \text{ cm}^2$. Find its radius.

- a. 6
- b. 24
- c. 12 ✓
- d. 72

$$r^2 = 144 \Rightarrow r = 12 \text{ cm}.$$

MCQ — Composite figures

73 Find the area of the L-shaped room shown (all measurements in meters).

- a. 70
- b. 58 ✓
- c. 46
- d. 82

$$\text{Full rectangle } 10 \times 7 = 70 \text{ minus the } 4 \times 3 \text{ notch} = 70 - 12 = 58 \text{ m}^2.$$

- 74 A garden is L-shaped: an 18 m by 8 m rectangle with a 6 m by 3 m rectangle removed from one corner. Find its area.

a. 108

b. 126 ✓

c. 162

d. 138

$$18 \times 8 = 144; 144 - 6 \times 3 = 144 - 18 = 126 \text{ m}^2.$$

- 75 Find the area of the shape shown: a 14 cm by 6 cm rectangle with a right triangle of base 6 cm and height 4 cm attached to one side.

a. 108

b. 90

c. 96 ✓

d. 72

$$\text{Rectangle} = 14 \times 6 = 84; \text{triangle} = \frac{1}{2}(6)(4) = 12; \text{total} = 96 \text{ cm}^2.$$

- 76 A rectangular lawn 20 m by 12 m has a circular flower bed of radius 3.5 m removed from it. Find the remaining lawn area, using $\pi = \frac{22}{7}$.

a. 196

b. 278.5

c. 240

d. 201.5 ✓

$$\text{Rectangle} = 240; \text{circle} = \frac{22}{7}(3.5)^2 = 38.5; \text{remaining} = 240 - 38.5 = 201.5 \text{ m}^2.$$

MCQ — Volume and surface area

77 Find the volume of the rectangular box shown.

a. 120 ✓

b. 60

c. 15

d. 30

$$V = 6 \times 5 \times 4 = 120 \text{ cm}^3.$$

78 Find the volume and surface area of the cube shown.

a. $V = 343 \text{ cm}^3$, $SA = 49 \text{ cm}^2$

b. $V = 49 \text{ cm}^3$, $SA = 294 \text{ cm}^2$

c. $V = 294 \text{ cm}^3$, $SA = 343 \text{ cm}^2$

d. $V = 343 \text{ cm}^3$, $SA = 294 \text{ cm}^2$ ✓

$$V = 7^3 = 343 \text{ cm}^3; SA = 6 \times 7^2 = 294 \text{ cm}^2.$$

79 Find the volume of the cylinder shown, using $\pi = \frac{22}{7}$.

a. 1,540 ✓

b. 770

c. 440

d. 3,080

$$V = \frac{22}{7}(7)^2(10) = 1,540 \text{ cm}^3.$$

80 Find the volume of the cone shown, using $\pi = \frac{22}{7}$.

a. 770 ✓

b. 2,310

c. 1,540

d. 385

$$V = \frac{1}{3} \times \frac{22}{7}(7)^2(15) = 770 \text{ cm}^3.$$

81 A rectangular prism has volume 360 cm^3 , length 12 cm, and width 5 cm. Find its height.

a. 6 ✓

b. 15

c. 72

d. 30

$$h = \frac{360}{12 \times 5} = 6 \text{ cm.}$$

MCQ — Coordinate geometry: distance and midpoint

82 Find the distance between the two points shown.

a. 5 ✓

b. 25

c. 7

d. $\sqrt{7}$

$$\sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9+16} = \sqrt{25} = 5.$$

83 Find the midpoint of the segment joining the two points shown.

a. (6, 3)

b. (3, 5)

c. (10, 6)

d. (5, 3) ✓

$$\text{Midpoint} = \left(\frac{2+8}{2}, \frac{3+3}{2} \right) = (5, 3).$$

84 Find the distance between the points (0, 0) and (6, 8).

a. 14

b. 10 ✓

c. 7

d. 48

$$\sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10.$$

85 Find the midpoint of the segment joining the two points shown.

a. (-1, -4)

b. (4, 1)

c. (2, 8)

d. (1, 4) ✓

$$\text{Midpoint} = \left(\frac{-3 + 5}{2}, \frac{1 + 7}{2} \right) = (1, 4).$$

MCQ — Similar figures and scale factor

86 Two similar triangles have a scale factor of 3 : 5. If the smaller triangle's perimeter is 24 cm, find the larger triangle's perimeter.

a. 14.4

b. 72

c. 64

d. 40 ✓

$$24 \times \frac{5}{3} = 40 \text{ cm.}$$

87 A model car is built at a scale of 1 : 24. If the model is 8 cm long, find the actual car's length in meters.

- a. 19.2
- b. 0.192
- c. 1.92 ✓
- d. 24

$$8 \times 24 = 192 \text{ cm} = 1.92 \text{ m.}$$

88 Two similar rectangles have areas in ratio 4 : 9. Find the ratio of their side lengths.

- a. 4 : 9
- b. 2 : 3 ✓
- c. 8 : 18
- d. 16 : 81

$$\text{Side ratio} = \sqrt{4} : \sqrt{9} = 2 : 3.$$

89 A photo is enlarged by a scale factor of 2.5. If the original photo is 6 cm wide, find the width of the enlargement.

- a. 8.5
- b. 12.5
- c. 3.6
- d. 15 ✓

$$6 \times 2.5 = 15 \text{ cm.}$$

MCQ — Word problems

- 90 A rectangular pool measuring 25 m by 12 m is surrounded by a deck 2 m wide. Find the area of the deck alone.

- a. 74
- b. 300
- c. 164 ✓
- d. 464

$$\text{Outer} = 29 \times 16 = 464; \text{pool} = 300; \text{deck} = 464 - 300 = 164 \text{ m}^2.$$

- 91 Tiling costs 38 riyals per square meter. Find the total cost to tile a rectangular room measuring 9 m by 6 m.

- a. 1,026 SAR
- b. 4,104 SAR
- c. 342 SAR
- d. 2,052 SAR ✓

$$\text{Area} = 9 \times 6 = 54 \text{ m}^2; \text{cost} = 54 \times 38 = 2,052 \text{ SAR}.$$

- 92 A water tank shaped like the rectangular box shown has these dimensions in meters. Find its capacity in cubic meters.

- a. 150
- b. 60
- c. 300 ✓
- d. 21

$$V = 10 \times 6 \times 5 = 300 \text{ m}^3.$$

Multiple choice — easy

93 A right triangle has legs 11 cm and 60 cm. Find the hypotenuse.

a. 61 ✓

b. 59

c. 71

d. 63

$$c = \sqrt{11^2 + 60^2} = \sqrt{121 + 3600} = \sqrt{3721} = 61 \text{ cm.}$$

94 A right triangle has legs 44 cm and 240 cm. Find the hypotenuse.

a. 244 ✓

b. 242

c. 246

d. 284

$$c = \sqrt{44^2 + 240^2} = \sqrt{1936 + 57600} = \sqrt{59536} = 244 \text{ cm.}$$

95 A right triangle has legs 5 cm and 12 cm. Find the hypotenuse.

a. 15

b. 11

c. 13 ✓

d. 17

$$c = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13 \text{ cm.}$$

96 A right triangle has legs 33 cm and 180 cm. Find the hypotenuse.

a. 183 ✓

b. 181

c. 213

d. 185

$$c = \sqrt{33^2 + 180^2} = \sqrt{1089 + 32400} = \sqrt{33489} = 183 \text{ cm.}$$

97 A right triangle has legs 10 cm and 24 cm. Find the hypotenuse.

- a. 34
- b. 24
- c. 28
- d. 26 ✓

$$c = \sqrt{10^2 + 24^2} = \sqrt{100 + 576} = \sqrt{676} = 26 \text{ cm.}$$

98 A right triangle has legs 12 cm and 16 cm. Find the hypotenuse.

- a. 18
- b. 20 ✓
- c. 28
- d. 22

$$c = \sqrt{12^2 + 16^2} = \sqrt{144 + 256} = \sqrt{400} = 20 \text{ cm.}$$

99 A right triangle has legs 36 cm and 160 cm. Find the hypotenuse.

- a. 196
- b. 164 ✓
- c. 162
- d. 166

$$c = \sqrt{36^2 + 160^2} = \sqrt{1296 + 25600} = \sqrt{26896} = 164 \text{ cm.}$$

100 A right triangle has legs 20 cm and 48 cm. Find the hypotenuse.

- a. 54
- b. 52 ✓
- c. 50
- d. 68

$$c = \sqrt{20^2 + 48^2} = \sqrt{400 + 2304} = \sqrt{2704} = 52 \text{ cm.}$$

101 A right triangle has legs 3 cm and 4 cm. Find the hypotenuse.

- a. 3
- b. 9
- c. 7
- d. 5 ✓

$$c = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5 \text{ cm.}$$

102 A right triangle has legs 9 cm and 40 cm. Find the hypotenuse.

- a. 39
- b. 41 ✓
- c. 43
- d. 49

$$c = \sqrt{9^2 + 40^2} = \sqrt{81 + 1600} = \sqrt{1681} = 41 \text{ cm.}$$

103 A right triangle has legs 21 cm and 72 cm. Find the hypotenuse.

- a. 77
- b. 93
- c. 75 ✓
- d. 73

$$c = \sqrt{21^2 + 72^2} = \sqrt{441 + 5184} = \sqrt{5625} = 75 \text{ cm.}$$

104 A right triangle has legs 12 cm and 35 cm. Find the hypotenuse.

a. 37 ✓

b. 47

c. 39

d. 35

$$c = \sqrt{12^2 + 35^2} = \sqrt{144 + 1225} = \sqrt{1369} = 37 \text{ cm.}$$

105 A right triangle has legs 27 cm and 120 cm. Find the hypotenuse.

a. 121

b. 125

c. 123 ✓

d. 147

$$c = \sqrt{27^2 + 120^2} = \sqrt{729 + 14400} = \sqrt{15129} = 123 \text{ cm.}$$

106 A right triangle has legs 24 cm and 45 cm. Find the hypotenuse.

a. 49

b. 53

c. 51 ✓

d. 69

$$c = \sqrt{24^2 + 45^2} = \sqrt{576 + 2025} = \sqrt{2601} = 51 \text{ cm.}$$

107 A right triangle has legs 48 cm and 140 cm. Find the hypotenuse.

- a. 146
- b. 150
- c. 188
- d. 148 ✓

$$c = \sqrt{48^2 + 140^2} = \sqrt{2304 + 19600} = \sqrt{21904} = 148 \text{ cm.}$$

Multiple choice — medium

108 A rectangle has length 22 cm and width 21 cm. Find its perimeter and area.

- a. 86, 468
- b. 462, 86
- c. 90, 462
- d. 86, 462 ✓

$$\text{Perimeter} = 2(22 + 21) = 86 \text{ cm. Area} = 22 \times 21 = 462 \text{ cm}^2.$$

109 A rectangle has length 9 cm and width 8 cm. Find its perimeter and area.

- a. 34, 78
- b. 38, 72
- c. 34, 72 ✓
- d. 72, 34

$$\text{Perimeter} = 2(9 + 8) = 34 \text{ cm. Area} = 9 \times 8 = 72 \text{ cm}^2.$$

110 A rectangle has length 17 cm and width 6 cm. Find its perimeter and area.

a. 46, 102 ✓

b. 50, 102

c. 46, 108

d. 102, 46

Perimeter = $2(17 + 6) = 46$ cm. Area = $17 \times 6 = 102$ cm².

111 A rectangle has length 23 cm and width 5 cm. Find its perimeter and area.

a. 56, 121

b. 56, 115 ✓

c. 115, 56

d. 60, 115

Perimeter = $2(23 + 5) = 56$ cm. Area = $23 \times 5 = 115$ cm².

112 A rectangle has length 7 cm and width 5 cm. Find its perimeter and area.

a. 24, 41

b. 35, 24

c. 28, 35

d. 24, 35 ✓

Perimeter = $2(7 + 5) = 24$ cm. Area = $7 \times 5 = 35$ cm².

113 A rectangle has length 6 cm and width 4 cm. Find its perimeter and area.

a. 24, 24

b. 20, 24 ✓

c. 20, 30

d. 24, 20

Perimeter = $2(6 + 4) = 20$ cm. Area = $6 \times 4 = 24$ cm².

114 A rectangle has length 10 cm and width 4 cm. Find its perimeter and area.

- a. 28, 46
- b. 40, 28
- c. 28, 40 ✓
- d. 32, 40

$$\text{Perimeter} = 2(10 + 4) = 28 \text{ cm. Area} = 10 \times 4 = 40 \text{ cm}^2.$$

115 A rectangle has length 15 cm and width 4 cm. Find its perimeter and area.

- a. 42, 60
- b. 38, 66
- c. 60, 38
- d. 38, 60 ✓

$$\text{Perimeter} = 2(15 + 4) = 38 \text{ cm. Area} = 15 \times 4 = 60 \text{ cm}^2.$$

116 A rectangle has length 21 cm and width 3 cm. Find its perimeter and area.

- a. 52, 63
- b. 48, 69
- c. 48, 63 ✓
- d. 63, 48

$$\text{Perimeter} = 2(21 + 3) = 48 \text{ cm. Area} = 21 \times 3 = 63 \text{ cm}^2.$$

117 A rectangle has length 4 cm and width 3 cm. Find its perimeter and area.

a. 14, 12 ✓

b. 14, 18

c. 12, 14

d. 18, 12

Perimeter = $2(4 + 3) = 14$ cm. Area = $4 \times 3 = 12$ cm².

118 A rectangle has length 12 cm and width 6 cm. Find its perimeter and area.

a. 36, 72 ✓

b. 36, 78

c. 72, 36

d. 40, 72

Perimeter = $2(12 + 6) = 36$ cm. Area = $12 \times 6 = 72$ cm².

119 A rectangle has length 25 cm and width 13 cm. Find its perimeter and area.

a. 325, 76

b. 76, 331

c. 80, 325

d. 76, 325 ✓

Perimeter = $2(25 + 13) = 76$ cm. Area = $25 \times 13 = 325$ cm².

120 A rectangle has length 11 cm and width 5 cm. Find its perimeter and area.

a. 32, 61

b. 32, 55 ✓

c. 55, 32

d. 36, 55

Perimeter = $2(11 + 5) = 32$ cm. Area = $11 \times 5 = 55$ cm².

121 A rectangle has length 24 cm and width 17 cm. Find its perimeter and area.

- a. 86, 408
- b. 82, 414
- c. 408, 82
- d. 82, 408 ✓

$$\text{Perimeter} = 2(24 + 17) = 82 \text{ cm. Area} = 24 \times 17 = 408 \text{ cm}^2.$$

122 A rectangle has length 21 cm and width 14 cm. Find its perimeter and area.

- a. 294, 70
- b. 70, 294 ✓
- c. 70, 300
- d. 74, 294

$$\text{Perimeter} = 2(21 + 14) = 70 \text{ cm. Area} = 21 \times 14 = 294 \text{ cm}^2.$$

Multiple choice — hard

123 An isosceles triangle has two equal angles of 74° each. Find the third angle.

- a. 38
- b. 74
- c. 32 ✓
- d. 26

$$180 - 74 - 74 = 32^\circ.$$

124 An isosceles triangle has two equal angles of 59° each. Find the third angle.

- a. 59
- b. 62 ✓
- c. 68
- d. 56

$$180 - 59 - 59 = 62^\circ.$$

125 A triangle has two angles of 46° and 28° . Find the third angle.

- a. 74
- b. 106 ✓
- c. 111
- d. 101

$$180 - 46 - 28 = 106^\circ.$$

126 An isosceles triangle has two equal angles of 64° each. Find the third angle.

- a. 52 ✓
- b. 46
- c. 64
- d. 58

$$180 - 64 - 64 = 52^\circ.$$

127 A triangle has two angles of 92° and 56° . Find the third angle.

- a. 27
- b. 37
- c. 148
- d. 32 ✓

$$180 - 92 - 56 = 32^\circ.$$

128 Find the sum of the interior angles of a pentagon (5 sides).

- a. 360
- b. 540 ✓
- c. 720
- d. 900

$$\text{Sum} = (n - 2) \times 180^\circ = (5 - 2) \times 180^\circ = 540^\circ.$$

129 Find the sum of the interior angles of a decagon (10 sides).

- a. 1620
- b. 1440 ✓
- c. 1260
- d. 1800

$$\text{Sum} = (n - 2) \times 180^\circ = (10 - 2) \times 180^\circ = 1440^\circ.$$

130 Find the sum of the interior angles of a nonagon (9 sides).

- a. 1440
- b. 1080
- c. 1260 ✓
- d. 1620

$$\text{Sum} = (n - 2) \times 180^\circ = (9 - 2) \times 180^\circ = 1260^\circ.$$

131 Find the sum of the interior angles of a heptagon (7 sides).

a. 1260

b. 900 ✓

c. 720

d. 1080

$$\text{Sum} = (n - 2) \times 180^\circ = (7 - 2) \times 180^\circ = 900^\circ.$$

132 Find the sum of the interior angles of an octagon (8 sides).

a. 1080 ✓

b. 1440

c. 1260

d. 900

$$\text{Sum} = (n - 2) \times 180^\circ = (8 - 2) \times 180^\circ = 1080^\circ.$$

133 A circle has radius 49 cm. Find its area, using $\pi = \frac{22}{7}$.

a. 7532

b. 308

c. 7546 ✓

d. 7560

$$A = \pi r^2 = \frac{22}{7} \times 2401 = 7546 \text{ cm}^2.$$

134 A circle has radius 28 cm. Find its area, using $\pi = \frac{22}{7}$.

a. 2450

b. 2478

c. 176

d. 2464 ✓

$$A = \pi r^2 = \frac{22}{7} \times 784 = 2464 \text{ cm}^2.$$

135 A circle has radius 14 cm. Find its circumference, using $\pi = \frac{22}{7}$.

- a. 84
- b. 616
- c. 88 ✓
- d. 92

$$C = 2\pi r = 2 \times \frac{22}{7} \times 14 = 88 \text{ cm.}$$

136 A circle has radius 35 cm. Find its area, using $\pi = \frac{22}{7}$.

- a. 3864
- b. 220
- c. 3850 ✓
- d. 3836

$$A = \pi r^2 = \frac{22}{7} \times 1225 = 3850 \text{ cm}^2.$$

137 A circle has radius 14 cm. Find its area, using $\pi = \frac{22}{7}$.

- a. 616 ✓
- b. 630
- c. 602
- d. 88

$$A = \pi r^2 = \frac{22}{7} \times 196 = 616 \text{ cm}^2.$$